# Probabilistic Modelling: Simulating Asset returns using Monte Carlo methods

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November 2025

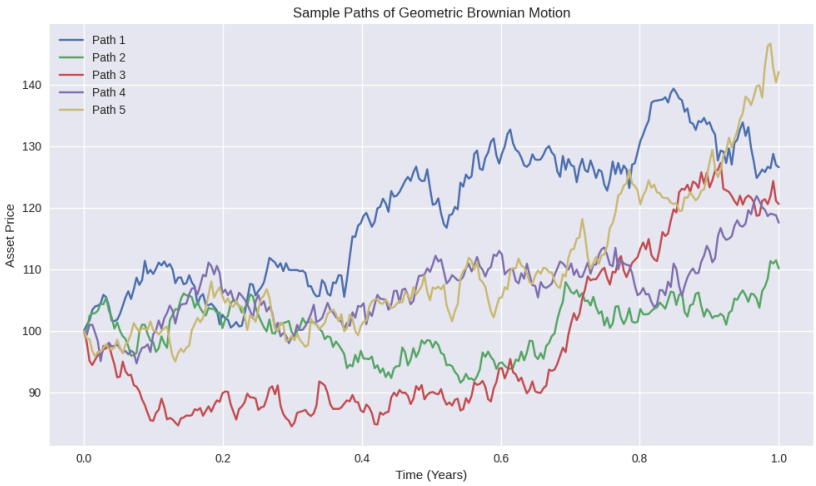
# Monte Carlo Option Pricing

## What is it?

Monte Carlo is an algorithm that uses random samples to obtain results. The underlying idea is to use the randomness of the variables to model many possible future paths of the asset's future price.

Core Ideas Include:

- Risk-neutral valuation: the option price equals the discounted expected payoff under risk-neutral probabilities
- Simulation of paths: where thousands (or even millions) of possible asset price trajectories are generated using stochastic models (having a random probability) like Geometric Brownian Motion (GBM).
- **Geometric Brownian Motion**: **(GBM)** is the mathematical model that says stock prices grow exponentially on average but fluctuate randomly, making it the standard tool for simulating asset paths in option pricing,




The above chart shows the Brownian Motion, each line represents a future price trajectory of the asset. Notice how each line starts at the same initial value of 100, but quickly diverge due to the random shocks embedded in the model. These random zig-zag movements mimic real-world volatility. Despite the randomness, each path exhibits an upward trend over time, driven by the model's drift component. The Brownian Motion captures how assets move in real time- just like the s&p 500, it grows on average but the exact path is uncertain.

GBM is significant as it is one of the stochastic models used to generate each simulated path for a Monte Carlo statistical experiment.

Parameters:

The Stochastic differential equation for GBM is:

$$S_t = \mu S_t dt + \sigma S_t dW$$

Integrating this equation using Ito's Lemma, we get:

$$\ln\left(\frac{S_t}{S_0}\right) = \left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t$$

The GBM model for stock pricing is mentioned below:

$$S_t = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma\varepsilon\sqrt{\Delta t}}$$

where:

$S_t$ = stock price at time t
$S_0$ = initial stock price
μ = the expected return
$W_t$ = standard Brownian motion
ϵ = the standard normal random variable $N(0,1)$
$\Delta t$ = the elapsed time-period
σ = volatility of the asset (random shock used in simulation)

Let's explore the equation on a deeper analysis.

$$(\mu - \frac{1}{2}\sigma^2)t$$

This component of the equation represents the model's drift, adjusted by the standard drift-correction term. Here, μ denotes the expected rate of return, and multiplying it by $t$ reflects how that return accumulates over time. The subtraction of $\frac{1}{2}\sigma^2$ corrects for the fact that volatility pushes the average exponential growth upward. By removing half of the variance, the formula ensures that the expected price path stays aligned with the theoretical mean implied by geometric Brownian motion.

$$\sigma\varepsilon\sqrt{\Delta t}$$

This component captures the random, unpredictable component of the stock's movement. The expression $\sigma\varepsilon\sqrt{\Delta t}$ introduces volatility into the model: σ scales the randomness, and ε is a normally distributed random variable that injects the shock occurring over the small time interval $\Delta t$. Together, they represent the day-to-day fluctuations that cause actual prices to deviate from the smooth drift path.

**Thesis:**

How Monte Carlo simulation can be used to estimate the fair value of call options by modelling thousands of potential price paths

**How do we use it?**

**Excel:**

To implement risk-neutral Monte Carlo option pricing in excel, the theoretical model is discretized into a form compatible with spreadsheet calculations. Under the risk-neutral measure, the stock price evolves to:

$$S_T = S_0 exp((r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T} * Z)$$

Where:

$Z \sim N(0, 1)$ : Each Monte Carlo simulation corresponds to one realization of Z, generated in excel using the built-in random function

$S_0$: Initial price

$r$ : Risk-free rate

$\sigma$ : Volatility

$T$ : Time to Maturity (Days/252)

$Z$ : Random Shock (=NORM.S.INV(RAND())

$S_T$ : Terminal Price (Exponential Formula)

Each row of the simulation represents an independent realization of the terminal stock price $S_T$. The option payoff is calculated for each simulated path and discounted back to present value. The Monte Carlo estimate of the call option price is obtained by averaging these discounted payoffs across all simulations

$$C \approx \frac{1}{N}\sum_{i=1}^{N} e^{-rT} max(S_T^{(i)} - K, 0)$$

~ As the number of simulations $N$ increases, the estimated price converges to the theoretical fair value implied by risk-neutral valuation.

Excel Link:
https://uofc-my.sharepoint.com/:x:/r/personal/josh_tolentino_ucalgary_ca/Documents/Monte%20Carlo%20Simulation.xlsx?d=w079493d8cdd74f7593ecf50e746a4efe&csf=1&web=1&e=cToPdm

Assume:

Day to expire is 1 trading month (21 days)
Price and Strike price is $100 (varies with each stock)
Risk free rate is 2.5%
Historical Volatility of the option is 0.2

**Code:**

https://github.com/morgan-swe/Monte-Carlo-Simulation.git

## Convergence and Simulation Error

Monte Carlo simulation provides an approximate solution to the option pricing problem. The accuracy of a Monte Carlo estimate depends on the number of simulated price paths. As the number of simulations increases, the estimated option value converges to its theoretical fair price.

This convergence is guaranteed by the law of large numbers, which states that the sample average of discounted payoffs approaches the true expected value under the risk-neutral measure as the number of simulations $N$ becomes large.

However, Monte Carlo methods converge relatively slowly. The estimation error decreases at a rate proportional to

$$\vartheta\left(\frac{1}{\sqrt{N}}\right)$$

*This implies that achieving higher precision requires a substantial increase in the number of simulations. For example Reducing the estimation error by half requires approximately four times as many simulated paths.

## Drawbacks of Geometric Brownian Model for option pricing

Geometric Brownian Motion (GBM) is frequently applied in financial modeling to estimate stock price behaviour. However, despite its widespread use, the model has several important limitations when applied to real-world markets

1.) GBM assumes that both expected returns and volatility remain constant over time. In practice, these variables fluctuate significantly in response to economic cycles, market uncertainty, and major news events. Treating them as fixed parameters can lead to inaccurate forecasts, especially during periods of heightened market instability.

2.) The model is built on the assumption of a perfectly efficiency and frictionless market. Real financial markets deviate from the ideal due to inefficiencies, information asymmetry, transaction costs, and behavioural biases, all of which undermine the realism of the assumption

3.) GBM does not explicitly incorporate interest rate dynamics. Central bank policy decisions, changes in interest rates play a critical role in shaping market behaviour. Ignoring this factor can result in predicted prices that fail to reflect actual market conditions

**Notes:**

**Call Option:** Standardized Contract that gives the buyer the right, but not the requirement to buy or sell a stock at a specified price for a specified amount of time.

- If a stock price is trading above the strike price, there will be some value. (For a Call option)
- If a stock price is trading below the strike price, there will be no value. (For a  Call option)
- If a stock price is trading below the strike price, the difference between the stock price and the strike price will be the value (For a Put Option)

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